Can You Taco Tongue and Evil Eyebrow?

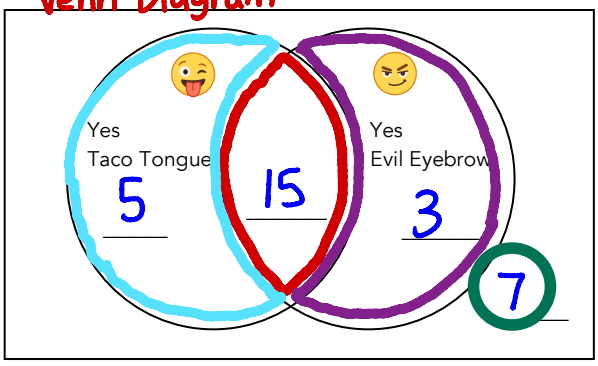
Some people believe that the ability to taco tongue and evil eyebrow is something that you are born with. Is this true? Are the two abilities somehow related?

1. Collect class data and complete the following.

Two-way table

	Yes Evil Eyebrow	No Evil Eyebrow	Total
Yes Taco Tongue	15	5	20
No Taco Tongue	3	7	10
Total	18	12	30

Venn Diagram



2. Suppose that we randomly choose a student from class. Find the following probabilities as fractions.

P(Yes Taco Tongue) = $20/30$

P(Yes Evil Eyebrow) = $18/30$

P(No Taco Tongue) = $10/30$

P(No Evil Eyebrow) = $12/30$

P(Yes Taco Tongue AND Yes Evil Eyebrow) = $15/30$

P(Yes Evil Eyebrow AND No Taco Tongue) = $3/30$

P(Yes Taco Tongue AND No Evil Eyebrow) = $5/30$

P(No Taco Tongue AND No Evil Eyebrow) = $7/30$

3. Suppose that we randomly choose a student from class. Find the following probabilities as fractions.

Mutually exclusive events

P(Yes Evil Eyebrow) = $18/30$

P(No Evil Eyebrow) = $12/30$

P(Yes Evil Eyebrow OR No Evil Eyebrow) = $30/30 = 18/30 + 12/30$

$P(A \text{ or } B) = P(A) + P(B)$

4. Suppose that we randomly choose a student from class. Find the following probabilities as fractions.

Not mutually exclusive events

P(Yes Taco Tongue) = $20/30$

P(Yes Evil Eyebrow) = $18/30$

P(Yes Taco Tongue OR Yes Evil Eyebrow) = $20/30 + 18/30 = 38/30 - 15/30 = 23/30$

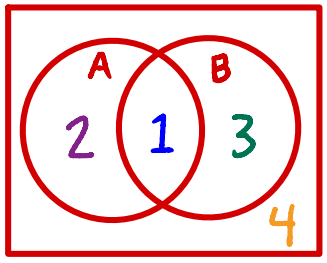
Addition Rule → $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$

Lesson 3.4 – The Addition Rule

QuickNotes

LT#1 Two way tables/Venn Diagram

	B	B^c
A	1	2
A^c	3	4



LT#2 Addition Rule

$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$

$A \cup B$ "union" "intersection" $A \cap B$

If events A and B are mutually exclusive, they can't occur together

$P(A \text{ and } B) = 0$, so $P(A \text{ or } B) = P(A) + P(B)$

Check Your Understanding

A random sample of high school students was surveyed regarding their procrastination habits. The two-way table displays the results. Suppose we choose a member of the sample at random.

Define events X: yes to taking an AP class and Y: yes to procrastinating often.

		Taking an AP class?		
		Yes	No	
Procrastinate often?	Yes	58	24	82
	No	12	22	34
		70	46	116

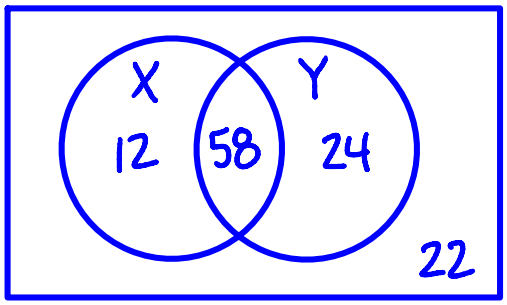
a. Explain in plain language what $P(Y^C)$ means and find the probability.

The probability that the student selected does not procrastinate often. $P(Y^c) = 34/116 = 0.293$.

b. Explain why $P(X \text{ or } Y) \neq P(X) + P(Y)$. Then find $P(X \text{ or } Y)$.

The events "yes to taking an AP class" and "yes to procrastinating often" are not mutually exclusive. $\frac{70}{116} + \frac{82}{116} - \frac{58}{116} = \frac{94}{116} = 0.81$

c. Make a Venn diagram to display the results.



d. Write the event "Does not procrastinate often and is taking an AP class" in symbolic form and find the probability.

$P(Y^c \text{ and } X) = 12/116 = 0.103$